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(54) **COAXIAL SPEAKER BOX**

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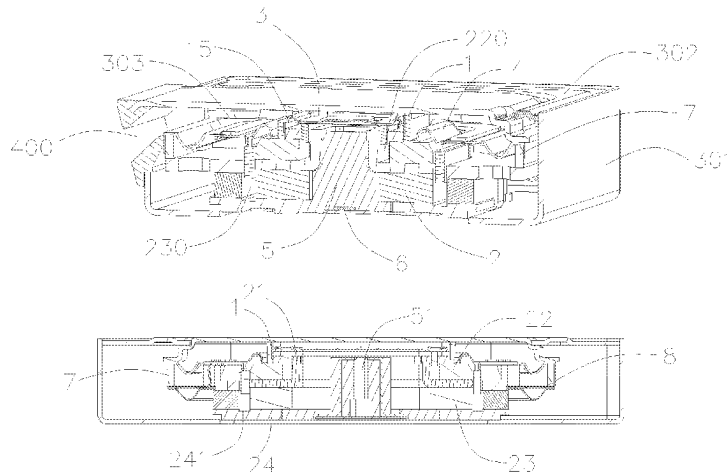
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(57) **ABSTRACT**

A coaxial speaker box includes a housing, a speaker, and a sound guide channel for guiding sound outside. The speaker includes a frame, a magnetic circuit system, a first vibration system, and a second vibration system. The first vibration system is disposed in the frame. The second vibration system is disposed on an outer side of the frame. A front cavity is formed among the first vibration system, the second vibration system, and the housing. The first vibration system includes a first vibrating diaphragm and a first voice coil. A coaxial full-band speaker is formed by the bass and the high pitch of the coaxial speaker box to provide high-quality sound effects. The high pitch and the bass share a side sound guide channel, and form a front cavity resonance in a bass channel to improve high-frequency sound effects.

9 Claims, 5 Drawing Sheets



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2460/13; H04R 1/227; H04R 9/10; H04R
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See application file for complete search history.

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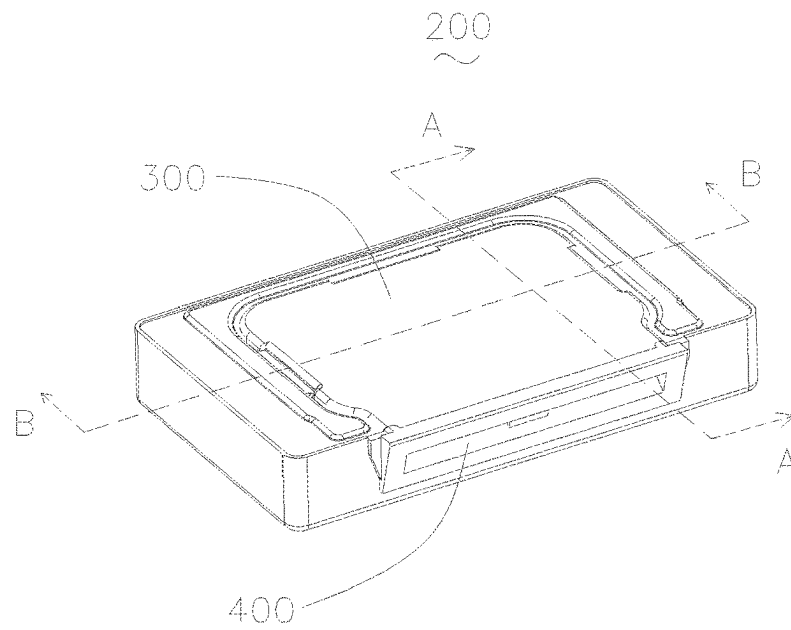


FIG. 1

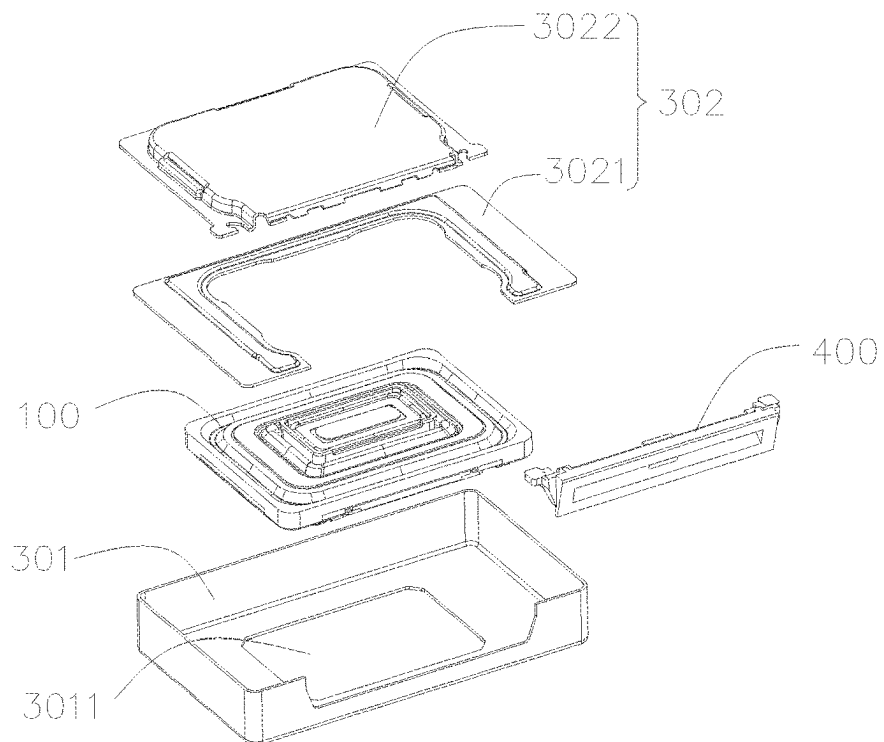


FIG. 2

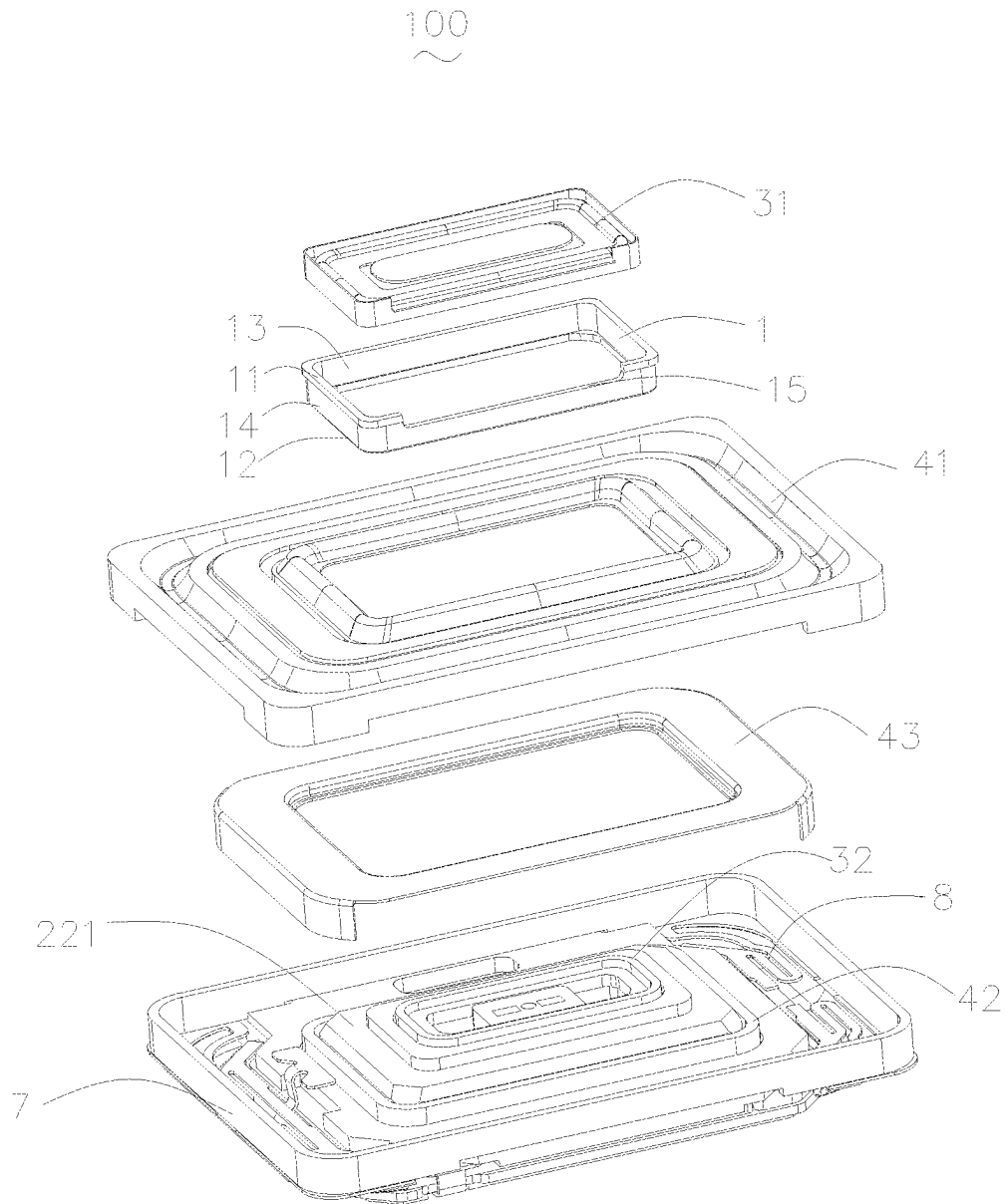


FIG. 3

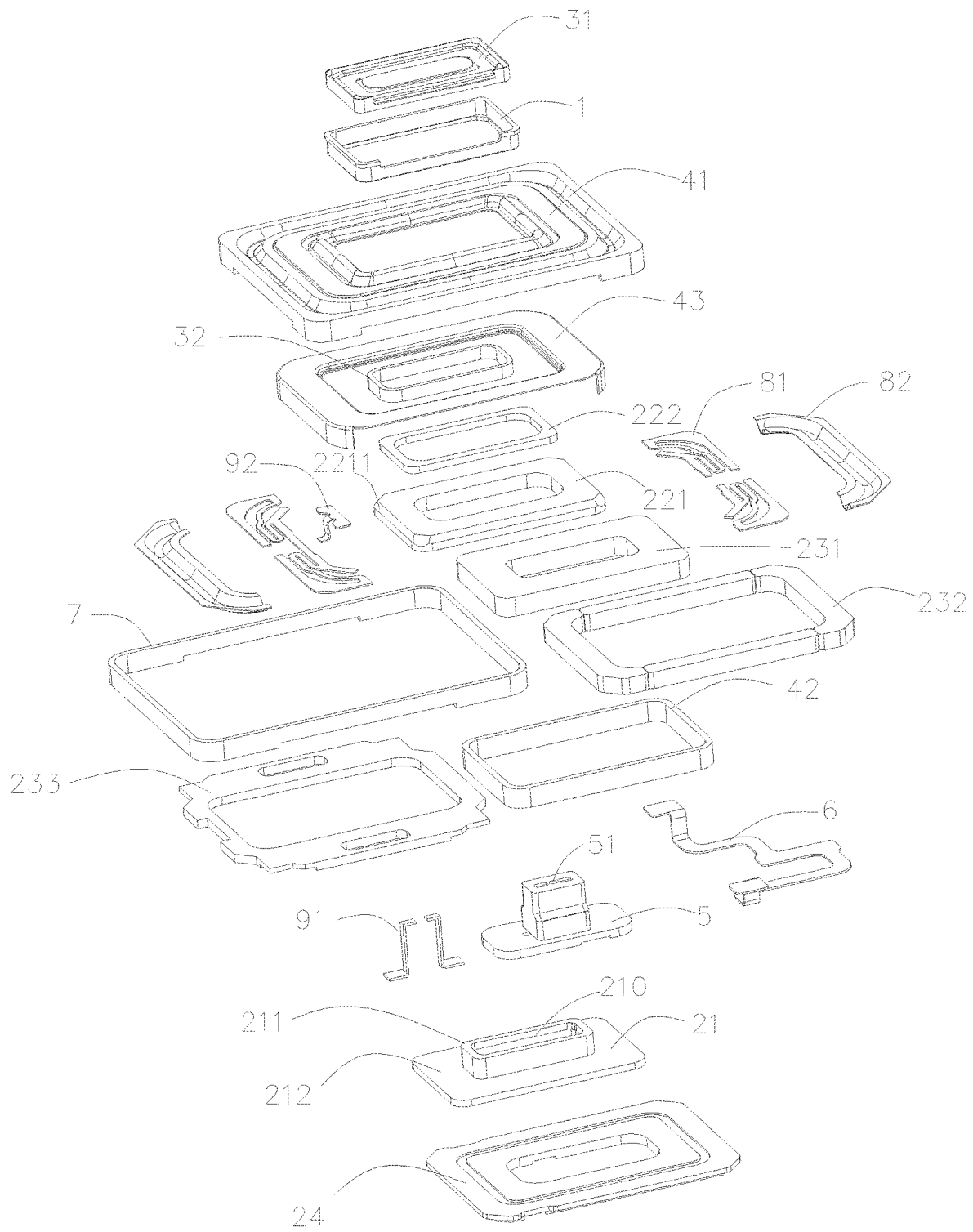


FIG. 4

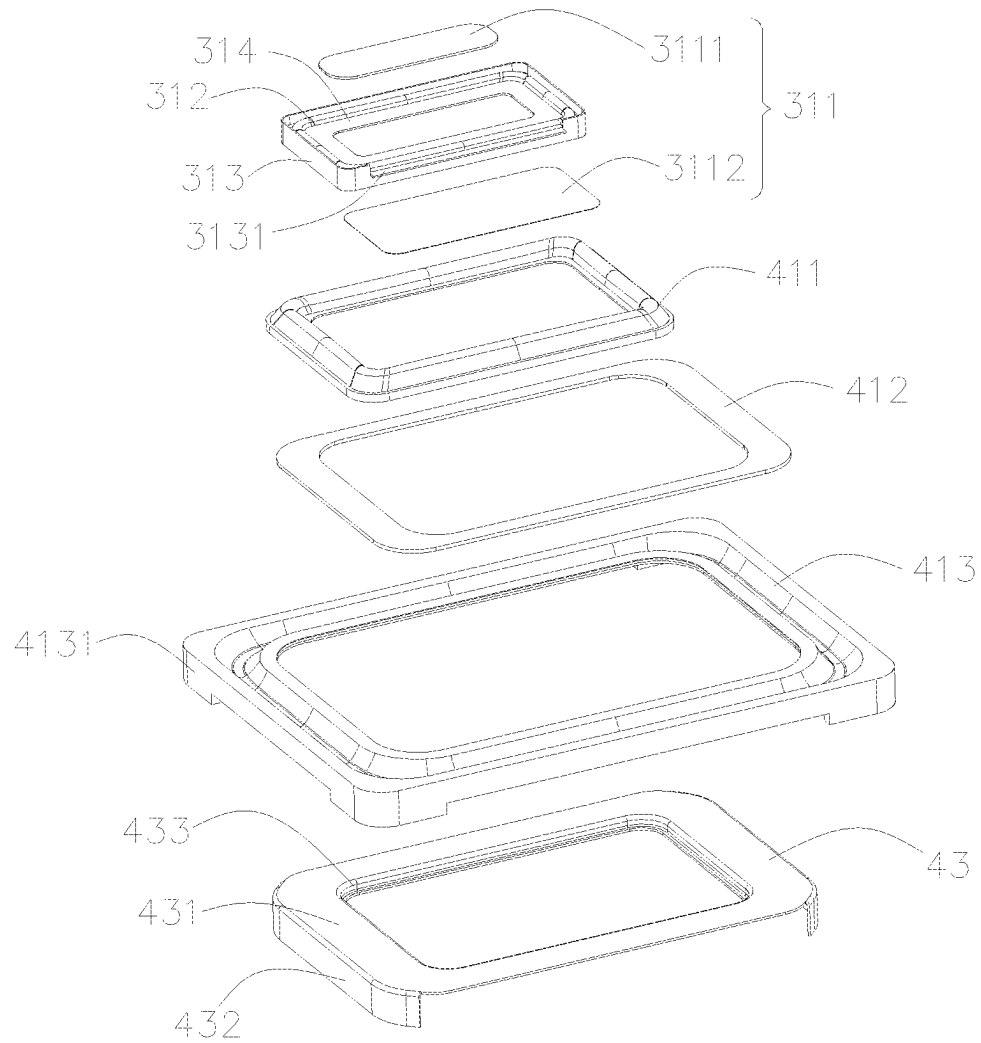


FIG. 5

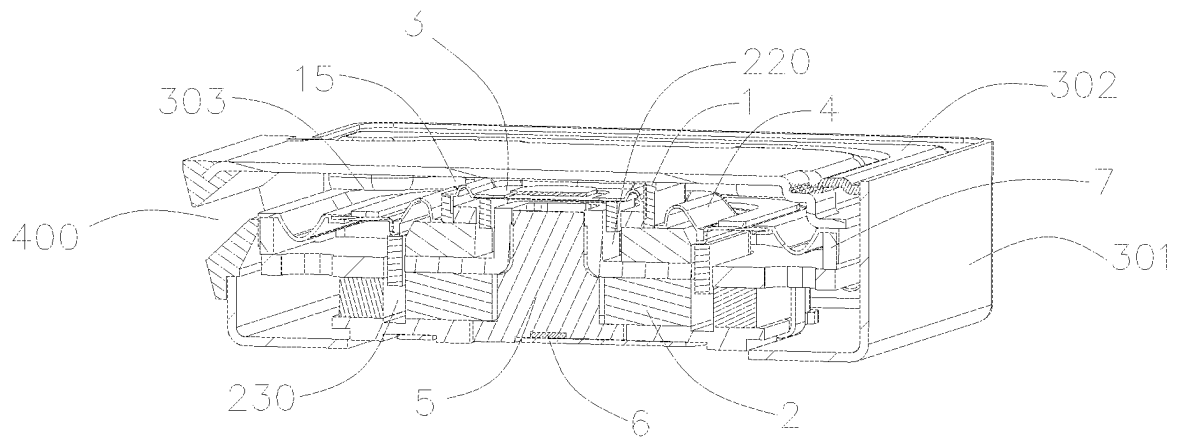


FIG. 6

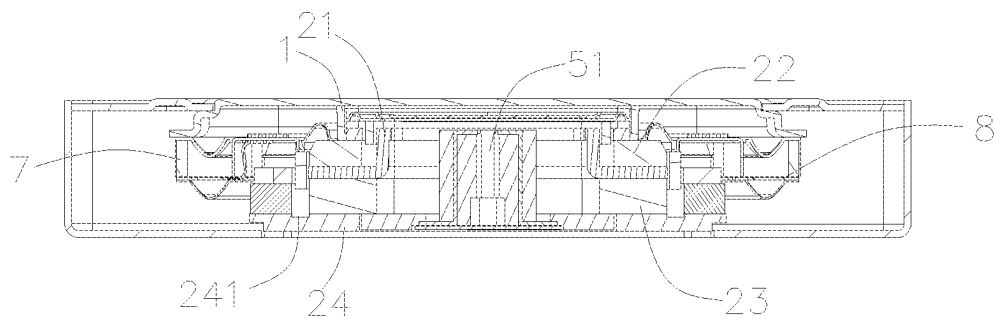


FIG. 7

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COAXIAL SPEAKER BOX**TECHNICAL FIELD**

The present disclosure relates to a field of sensors, and in particular to a coaxial speaker box.

BACKGROUND

Coaxial speakers refer to speakers each integrated with a tweeter speaker and a woofer speaker, where the tweeter speaker and the woofer speaker are coaxially disposed. A sound source of the tweeter speaker of the coaxial speakers and a sound source of the woofer speaker of the coaxial speakers are disposed on a front surface of coaxial speaker boxes. That is, sound production directions of the tweeter speaker and the woofer speaker are the same. In addition, the tweeter speaker and the woofer speaker may be further respectively disposed on the front surface and a back surface of the coaxial speaker box. That is, the sound production directions of the tweeter speaker and the woofer speaker are different. Each of the coaxial speakers is assembled into a housing to form a coaxial speaker box.

The tweeter speaker and the woofer speaker of the coaxial speakers in related art respectively include a separated vibration system and a separated magnetic circuit system, which is complex in structure. A whole height of the coaxial speakers and the coaxial speaker boxes are increased, where the coaxial speakers are composed of the tweeter speaker and the woofer speaker, so that inconvenience of installation is resulted and applicability of products is reduced. In addition, a separated high pitch magnetic circuit and a separated bass magnetic circuit further increase production costs, which reduces market competitiveness of the products. What's more, in related art, an effective area of a vibrating diaphragm in a speaker, distal from a sound surface, is limited, which cannot meet requirements of full-sound range. Meanwhile, a sound production direction of the coaxial speaker boxes having the coaxial speakers is generally at top, so that resonance energy of a front cavity is low and high pitch sound effect is poor.

Therefore, it is necessary to provide a new coaxial speaker box to solve above technical problems.

SUMMARY

The present disclosure aims to provide a full-band coaxial speaker box with good high pitch and bass sound effects.

In order to achieve above purpose, the present disclosure provides the coaxial speaker box, including a housing, a speaker, and a sound guide channel. The speaker is accommodated in the housing. The sound guide channel is configured to guide sound emitted by the speaker from a side surface of the housing to an outside of the housing. The speaker includes a frame, a magnetic circuit system, a first vibration system, and a second vibration system. The magnetic circuit system is fixed below the frame. The first vibration system is fixed to an inner side of the frame. The second vibration system is fixed to an outer side of the frame. The second vibration system is disposed around the first vibration system. The first vibration system and the second vibration system are coaxially disposed. A front cavity is formed among the first vibration system, the second vibration system, and the housing. The first vibration system includes a first vibrating diaphragm and a first voice coil. The first voice coil is configured to drive the first vibrating diaphragm to vibrate and generate sound. The second vibra-

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tion system includes a second vibrating diaphragm and a second voice coil. The second voice coil is configured to drive the second vibrating diaphragm to vibrate and generate sound. The magnetic circuit system includes a first magnetic gap and a second magnetic gap. The second magnetic gap is formed on an outer side of the first magnetic gap. The first voice coil is at least partially inserted into the first magnetic gap. The second voice coil is at least partially inserted into the second magnetic gap. The first vibrating diaphragm is for high-pitch sounding. The high pitch emitted by the first vibrating diaphragm penetrates through the frame and is transmitted to an outside of the housing from the sound guide channel. The second diaphragm is for bass sounding. The bass emitted by the second vibrating diaphragm is transmitted to the outside of the housing from the sound guide channel.

Furthermore, the frame is of an annular structure. The frame includes an upper wall surface, a lower wall surface, an inner wall surface, and an outer wall surface. The upper wall surface abuts against the housing. The lower wall surface is fixed on the magnetic circuit system. The inner wall surface is connected to the upper wall surface and the lower wall surface. The frame is recessed from the upper wall surface towards the lower wall surface to form a first sound transmission hole. The first sound transmission hole does not penetrate to the lower wall surface.

Furthermore, a length of the sound guide channel is greater than a length of the first sound transmission hole of the frame.

Furthermore, the speaker further includes an insert and a first circuit board. The insert is disposed around the magnetic circuit system. The first circuit board is fixed with the insert. The insert includes a first terminal. The first terminal penetrates through the insert. The first voice coil is electrically connected to the first circuit board through the first terminal.

Furthermore, the speaker further includes a supporting frame and an elastic assembly. The elastic assembly is connected to the second vibration system and the supporting frame. The elastic assembly includes a second circuit board and a supporting vibrating diaphragm. The supporting vibrating diaphragm is connected to the second circuit board. The second voice coil is electrically connected to the second circuit board.

Furthermore, the magnetic circuit system includes a magnetic bowl. The magnetic bowl includes a side portion and a bottom portion. The side portion is enclosed to form a containing space. The bottom portion is formed by bending and extending from one end of the side portion. The magnetic circuit system further includes a first magnet assembly and a second magnet assembly. The first magnet assembly is disposed on a first side of the magnetic bowl. The second magnet assembly is disposed on a second side of the magnetic bowl. The first magnet assembly includes a first magnet. The first magnetic gap is formed between the first magnet and a side wall of the magnetic bowl. The second magnet assembly includes a second magnet and a third magnetic steel. The third magnetic steel is disposed on an outer side of the second magnet. The third magnetic steel and the second magnet form the second magnetic gap.

Furthermore, the first vibrating diaphragm includes a first dome and a first folding ring. The first folding ring surrounds the first dome. The first folding ring includes a first fixing portion. The first fixing portion is attached and fixed to the inner wall surface of the frame.

Furthermore, a second sound transmission hole is defined in a position, with respect to the first sound transmission hole of the frame, of the first fixing portion.

Furthermore, parts of the first fixing portion forming the second sound transmission hole are attached and fixed to a bottom wall of the frame, where the first sound transmission hole is defined on the bottom wall of the frame.

Furthermore, the first magnet assembly includes a pole core. The pole core is disposed on the first magnet. The first fixing portion is partially clamped between an inner wall surface of the frame and the pole core.

Compared with prior art, the coaxial speaker box of the present disclosure includes: the housing, the speaker, and the sound guide channel. The speaker is accommodated in the housing. The sound guide channel is configured to guide sound emitted by the speaker from the side surface of the housing to the outside of the housing. The speaker includes the frame, the magnetic circuit system, the first vibration system, and the second vibration system. The magnetic circuit system is fixed below the frame. The first vibration system is fixed to the inner side of the frame. The second vibration system is fixed to the outer side of the frame. The second vibration system is disposed around the first vibration system. The first vibration system and the second vibration system are coaxially disposed. The front cavity is formed among the first vibration system, the second vibration system, and the housing. The first vibration system includes the first vibrating diaphragm and the first voice coil. The first voice coil is configured to drive the first vibrating diaphragm to vibrate and generate sound. The second vibration system includes the second vibrating diaphragm and the second voice coil. The second voice coil is configured to drive the second vibrating diaphragm to vibrate and generate sound. The magnetic circuit system includes the first magnetic gap and the second magnetic gap. The second magnetic gap is formed on an outer side of the first magnetic gap. The first voice coil is at least partially inserted into the first magnetic gap. The second voice coil is at least partially inserted into the second magnetic gap. The first vibrating diaphragm is for high-pitch sounding. The high pitch emitted by the first vibrating diaphragm penetrates through the frame and is transmitted to an outside of the housing from the sound guide channel. The second diaphragm is for bass sounding. The bass emitted by the second vibrating diaphragm is transmitted to the outside of the housing from the sound guide channel. The coaxial speaker box of the present disclosure provides an annular bass vibration system to provide bass sound effects. A high pitch vibration system is disposed on an inner side of the bass vibration system to provide high pitch sound effects. A coaxial full-band speaker is formed by bass and high pitch to provide high-quality sound effects for the coaxial speaker box. The high pitch and the bass share a side sound guide channel, and form a front cavity resonance in a bass channel to improve high-frequency sound effects. high pitch airflow emits sound from one side of the frame, which improves directivity of the coaxial speaker box.

BRIEF DESCRIPTION OF DRAWINGS

In order to more clearly illustrate technical solutions in embodiments of the present disclosure, drawings required in description of the embodiments are briefly described below. Obviously, the drawings in the following description are merely some embodiments of the present disclosure. For a person of ordinary skill in art, other drawings may be obtained according to the drawings without creative efforts.

FIG. 1 is a three-dimensional structural schematic diagram of a coaxial speaker box according to one embodiment of the present disclosure.

FIG. 2 is a partially exploded schematic diagram of the coaxial speaker box as shown in FIG. 1.

FIG. 3 is a schematic diagram of a partially exploded view of a speaker in the coaxial speaker box as shown in FIG. 1.

FIG. 4 is a schematic diagram of another partially exploded view of the speaker in the coaxial speaker box as shown in FIG. 1.

FIG. 5 is a schematic diagram of an exploded view of partially components of the speaker in the coaxial speaker box as shown in FIG. 1.

FIG. 6 is a cross-sectional schematic diagram taken along the line A-A shown in FIG. 1.

FIG. 7 is a cross-sectional schematic diagram taken along the line B-B shown in FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS

Technical solutions in embodiments of the present disclosure are clearly and completely described below with reference to accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, not all of the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by a person of ordinary skill in art without creative efforts shall fall within a protection scope of the present disclosure.

Referring to FIGS. 1-7, the present disclosure provides a coaxial speaker box **200**, including a housing **300**, a speaker **100**, and a sound guide channel **400**. The speaker **100** is accommodated in the housing **300**. The sound guide channel **400** is configured to guide sound emitted by the speaker **100** from a side surface of the housing **300** to an outside of the housing **300**.

As shown in FIG. 2, the housing **300** includes a lower housing **301** and an upper housing **302**. The upper housing **302** covers the lower housing **301**. A through hole **3011** is defined in a bottom wall of the lower housing **301**. The upper housing **302** includes a frame **3021** and a steel sheet **3022**. The steel sheet **3022** is embedded and fixed to the frame **3021**.

As shown in FIGS. 3-7, the speaker **100** includes a frame **1**, a magnetic circuit system **2**, a first vibration system **3**, a second vibration system **4**, an insert **5**, a first circuit board **6**, a supporting frame **7**, and an elastic assembly **8**. The magnetic circuit system **2** is fixed on the frame **1**. The first vibration system **3** is fixed on an inner side of the frame **1**. The second vibration system **4** is fixed on an outer side of the frame **1**. The first circuit board **6** is fixed on the insert **5**. The second vibration system **4** surrounds an outer side of the first vibration system **3**. The first vibration system **3** and the second vibration system **4** are coaxially disposed.

A periphery of the frame **1** is roughly rectangular. The frame **1** includes an upper wall surface **11**, a lower wall surface **12**, an inner wall surface **13**, and an outer wall surface **14**. The upper wall surface **11** abuts against the housing **300**. The lower wall surface **12** is fixed on the magnetic circuit system **2**. The inner wall surface **13** is connected to the upper wall surface **11** and the lower wall surface **12**. Specifically, the upper wall surface **11** abuts against and is fixed with the steel sheet **3022**. The frame **1** is recessed from the upper wall surface **11** towards the lower wall surface **12** to form a first sound transmission hole **15**. The first sound transmission hole **15** penetrates through the

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inner wall surface **13** and the outer wall surface **14**, and does not penetrate to the lower wall surface **12**. The first vibration system **3** is disposed on one side of the inner wall surface **13** of the frame **1**. The second vibration system **4** is disposed on one side of the outer wall surface **14** of the frame **1**. A length of the first sound transmission hole **15** of the frame **1** is smaller than a length of the sound guide channel **400**.

The magnetic circuit system **2** includes a magnetic bowl **21**, a first magnet assembly **22**, and a second magnet assembly **23**. The first magnet assembly **22** is disposed on a first side of the magnetic bowl **21**. The second magnet assembly **23** is disposed on a second side of the magnetic bowl **21**. The first magnet assembly **22** and the second magnet assembly **23** are coaxially disposed. The magnetic bowl **21** includes a side portion **211** and a bottom portion **212**. The side portion **211** is enclosed to form a containing space **210**. The bottom portion **212** is formed by bending and extending from one end of the side portion **211**. The first magnet assembly **22** includes a first magnet **221** and a pole core **222**. The pole core **222** is disposed on the first magnet **221**. A first magnetic gap **220** is defined between the first magnet **221** and the side portion **211** of the magnetic bowl **21**. A tangent plane **2211** is defined on an outer edge of a surface, close to the second vibration system **4**, of the first magnet **221**. The bottom portion **212** of the magnetic bowl **21** carries the first magnet **221**. The second magnet assembly **23** includes a second magnet **231**, a third magnetic steel **232**, and a magnetic conductive plate **233**. The third magnetic steel **232** and the second magnet **231** form a second magnetic gap **230**. The magnetic conductive plate **233** is stacked above the third magnetic steel **232**. The bottom portion **212** of the magnetic bowl **21** is stacked above the second magnet **231**. The first magnet **221** and the second magnet **232** are oppositely disposed. The magnetic circuit system **2** further includes a yoke **24** and the third magnetic steel **232**. The yoke **24** carries the second magnet **231**. The yoke **24** is partially exposed through a through hole **3011** of the lower housing **301**. A lower surface of the yoke **24** is flush with a lower surface of the insert **5**.

The first magnet **221**, the second magnet **231**, and the third magnetic steel **232** are magnetized in vibration direction of the speaker **100**. The first magnet **221** and the second magnet **231** are disposed opposite to each other in the same pole. That is, magnetizing directions of the first magnet **221** and the second magnet **231** are opposite. The second magnet **231** and the third magnetic steel **232** are opposite in magnetizing direction. In the embodiment, the first magnet **221**, the pole core **222**, the second magnet **231**, the third magnetic steel **232**, and the magnetic conductive plate **233** are all hollow annular structures.

The first vibration system **3** includes a first vibrating diaphragm **31** and a first voice coil **32**. The first voice coil **32** is configured to drive the first vibrating diaphragm **31** to vibrate and generate sound. The first voice coil **32** is at least partially inserted into the first magnetic gap **220**. The first vibrating diaphragm **31** is for high-pitch sounding, and a sounding direction of the first vibrating diaphragm **31** is upward along the vibration direction. The first vibrating diaphragm **31** includes a first dome **311** and a first folding ring **312**. The first folding ring **312** surrounds the first dome **311**. The first dome **311** includes a first dome portion **3111** and a second dome portion **3112**. The first dome portion **3111** is disposed above the first folding ring **312**. The second dome portion **3112** is disposed below the first folding ring **312**. The first folding ring **312** is of a hollow annular structure. The first folding ring **312** is of a structure protruding upwards in the vibration direction. A size of the first

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dome portion **3111** is smaller than a size of the second dome portion **3112**. The first dome portion **3111** is attached and fixed to the second dome portion **3112**. The second dome portion **3112** is fixed on the first folding ring **312**. A first fixing portion **313** is disposed on an outer edge of the first folding ring **312**. A second fixing portion **314** is disposed on an inner edge of the first folding ring **312**. The first fixing portion **313** is attached and fixed to an inner side wall **11** of the frame **1**. The second dome portion **3112** is attached to the second fixing portion **314**. The second dome portion **3112** is clamped between the second fixing portion **314** and the first voice coil **32**. Furthermore, the first fixing portion **313** is partially clamped between the frame **1** and the pole core **222**. A second sound transmission hole **3131** is defined in a position, with respect to the first sound transmission hole **15** of the frame **1**, of the first fixing portion **313**. Parts of the first fixing portion **313** forming the second sound transmission hole **3131** are attached and fixed to a bottom wall of the frame **1**, where the first sound transmission hole **15** is defined on the bottom wall of the frame **1**.

The second vibration system **4** includes a second vibrating diaphragm **41**, a second voice coil **42**, and a voice coil framework **43**. The second voice coil **42** is configured to drive the second vibrating diaphragm **41** to vibrate and generate sound. The voice coil framework **43** is connected to the second vibrating diaphragm **41** and the second voice coil **42**. The second voice coil **42** is at least partially inserted into the second magnetic gap **230**. The second diaphragm **41** is for bass sounding, and a sounding direction of the second diaphragm **41** is upward along the vibration direction. The second vibrating diaphragm **41** includes a second folding ring **411**, a second dome **412**, and a third folding ring **413**. The second dome **412** is disposed around the second folding ring **411**. The third folding ring **413** is disposed around the second dome **412**. The second folding ring **411**, the second dome **412**, and the third folding ring **413** are all hollow annular structures. The second folding ring **411** is of a structure protruding upwards in the vibration direction. The third folding ring **413** is of a structure which is sunken downwards in the vibration direction. The voice coil framework **43** is connected to the second folding ring **411**, the second dome **412**, the third folding ring **413**, and the second voice coil **42**. An end, close to the first vibrating diaphragm **31**, of the second folding ring **411** is fixed on the outer wall surface **14** or/and the first magnet **221** of the frame **1**. An outer edge of the third folding ring **413** includes a third fixing portion **4131**.

A front cavity **303** is formed among the first vibration system **3**, the second vibration system **4**, and the upper housing **302** of the housing **300**. A portion of the front cavity, with respect to the first vibration system **3**, is communicated with a portion of the front cavity, with respect to the second vibration system **4**, through the first sound transmission hole **15** and the second sound transmission hole **3131**. Due to a fact that a length of the sound guide channel **400** is larger than a length of the first sound transmission hole **15** of the frame **1**, a high pitch portion forms a horn shape along a gradual diffusion direction to be emitted to an outer side of the housing **300**, which improves directivity of the coaxial speaker box **200**, and improves resonance energy of the front cavity **303**.

A receding recess portion **241** is disposed on a position, with respect to the second voice coil **42**, of the yoke **24**, so as to provide a sufficient vibration space for the second voice coil **42**.

The insert **5** is disposed in a containing space **210** defined by the side portion **211** of the magnetic bowl **21**. The first

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circuit board 6 is attached to an outer surface of the yoke 24. The insert 5 includes a first terminal 91. The first terminal 91 penetrates through the insert 5. The first terminal 91 is connected to the first circuit board 6. The first voice coil 32 is electrically connected to the first circuit board 6 through the first terminal 91. The insert 5 includes a leakage channel 51. The leakage channel 51 penetrates through the insert 5.

The supporting frame 7 is an annular steel ring. The third fixing portion 4131 of the third folding ring 413 is fixedly connected to the supporting frame 7. Specifically, the third fixing portion 4131 covers an upper surface of the supporting frame 7 and part of an outer side surface of the supporting frame 7. The pole core 222 is fixed at one end, distal from the second vibrating diaphragm 41, of the supporting frame 7.

The elastic assembly 8 is connected to the voice coil framework 43 and the supporting frame 7. The elastic assembly 8 includes a second circuit board 81 and a supporting vibrating diaphragm 82. The supporting vibrating diaphragm 82 is connected to the second circuit board 81. The second voice coil 42 is electrically connected to the second circuit board 81. Specifically, the speaker 100 further includes a second terminal 92. The second terminal 92 is connected to the second circuit board 81. The second voice coil 42 is electrically connected to the second circuit board 81 through the second terminal 92. In the embodiment, the number of the elastic assemblies 8 is two, the elastic assemblies are disposed in a short-axis direction of the speaker 100, and the elastic assemblies are determined according to visual conditions in other embodiments.

The voice coil framework 43 includes a framework flat plate portion 431, a first extension portion 432, and a second extension portion 433. The framework flat plate portion 431 is attached to the second dome 412. The first extension portion 432 is bent and extending from a first end of the framework flat plate portion 431 towards the second circuit board 81. The first extension portion 432 is connected to the second circuit board 81. The second extension portion 433 is bent and extending from a second end of the framework flat plate portion 431 towards the second voice coil 42. The second extension portion 433 is connected to the second voice coil 42.

Compared with prior art, the coaxial speaker box of the present disclosure includes: the housing, the speaker, and the sound guide channel. The speaker is accommodated in the housing. The sound guide channel is configured to guide sound emitted by the speaker from the side surface of the housing to the outside of the housing. The speaker includes the frame, the magnetic circuit system, the first vibration system, and the second vibration system. The magnetic circuit system is fixed below the frame. The first vibration system is fixed to the inner side of the frame. The second vibration system is fixed to the outer side of the frame. The second vibration system is disposed around the first vibration system. The first vibration system and the second vibration system are coaxially disposed. The first vibration system includes the first vibrating diaphragm and the first voice coil. The first voice coil is configured to drive the first vibrating diaphragm to vibrate and generate sound. The second vibration system includes the second vibrating diaphragm and the second voice coil. The second voice coil is configured to drive the second vibrating diaphragm to vibrate and generate sound. The magnetic circuit system includes the first magnetic gap and the second magnetic gap. The second magnetic gap is formed on an outer side of the first magnetic gap. The first voice coil is at least partially inserted into the first magnetic gap. The second voice coil is

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at least partially inserted into the second magnetic gap. The first vibrating diaphragm is for high-pitch sounding. The high pitch emitted by the first vibrating diaphragm penetrates through the frame and is transmitted to an outside of the housing from the sound guide channel. The second diaphragm is for bass sounding. The bass emitted by the second vibrating diaphragm is transmitted to the outside of the housing from the sound guide channel. The coaxial speaker box of the present disclosure provides an annular bass vibration system to provide bass sound effects. A high pitch vibration system is disposed on an inner side of the bass vibration system to provide high pitch sound effects. A coaxial full-band speaker is formed by bass and high pitch to provide high-quality sound effects for the coaxial speaker box. The high pitch and the bass share a side sound guide channel, and form a front cavity resonance in a bass channel to improve high-frequency sound effects. High pitch airflow emits sound from one side of the frame, which improves directivity of the coaxial speaker box.

The above are only the embodiments of the present disclosure. It should be noted that, for the person of ordinary skill in the art, improvements are made without departing from concepts of the present disclosure, but these are all within the protection scope of the present disclosure.

What is claimed is:

1. A coaxial speaker box, comprising:

a housing;

a speaker; and

a sound guide channel;

wherein the speaker is accommodated in the housing, the sound guide channel is configured to guide sound emitted by the speaker from a side surface of the housing towards an outside of the housing; the speaker comprises a frame, a magnetic circuit system, a first vibration system, and a second vibration system; the magnetic circuit system is fixed below the frame, the first vibration system is fixed to an inner side of the frame, the second vibration system is fixed to an outer side of the frame, the second vibration system is disposed around the first vibration system; the first vibration system and the second vibration system are coaxially disposed, a front cavity is formed among the first vibration system, the second vibration system, and the housing; the first vibration system comprises a first vibrating diaphragm and a first voice coil, the first voice coil is configured to drive the first vibrating diaphragm to vibrate and generate sound; the second vibration system comprises a second vibrating diaphragm and a second voice coil, the second voice coil is configured to drive the second vibrating diaphragm to vibrate and generate sound; the magnetic circuit system comprises a first magnetic gap and a second magnetic gap, the second magnetic gap is formed on an outer side of the first magnetic gap; the first voice coil is at least partially inserted into the first magnetic gap, the second voice coil is at least partially inserted into the second magnetic gap, the first vibrating diaphragm is for high-pitch sounding, the high pitch emitted by the first vibrating diaphragm penetrates through the frame and is transmitted to an outside of the housing from the sound guide channel; the second diaphragm is for bass sounding, and the bass emitted by the second vibrating diaphragm is transmitted to the outside of the housing from the sound guide channel;

the speaker further comprises an insert and a first circuit board, the insert is surrounded by the magnetic circuit system, the first circuit board is fixed with the insert, the

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insert comprises a first terminal, the first terminal penetrates through the insert, and the first voice coil is electrically connected to the first circuit board through the first terminal.

2. The coaxial speaker box according to claim 1, wherein the frame is of an annular structure, the frame comprises an upper wall surface, a lower wall surface, an inner wall surface, and an outer wall surface; the upper wall surface abuts against the housing, the lower wall surface is fixed on the magnetic circuit system, the inner wall surface is connected to the upper wall surface and the lower wall surface, the frame is recessed from the upper wall surface towards the lower wall surface to form a first sound transmission hole, and the first sound transmission hole does not penetrate to the lower wall surface.

3. The coaxial speaker box according to claim 2, wherein a length of the sound guide channel is greater than a length of the first sound transmission hole of the frame.

4. The coaxial speaker box according to claim 1, wherein the speaker further comprises a supporting frame and an elastic assembly, the elastic assembly is connected to the second vibration system and the supporting frame, the elastic assembly comprises a second circuit board and a supporting vibrating diaphragm, the supporting vibrating diaphragm is connected to the second circuit board, and the second voice coil is electrically connected to the second circuit board.

5. The coaxial speaker box according to claim 4, wherein the magnetic circuit system comprises a magnetic bowl, the magnetic bowl comprises a side portion and a bottom portion, the side portion is enclosed to form a containing space, the bottom portion is formed by bending and extending from one end of the side portion; the magnetic circuit

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system further comprises a first magnet assembly and a second magnet assembly, the first magnet assembly is disposed on a first side of the magnetic bowl, the second magnet assembly is disposed on a second side of the magnetic bowl, the first magnet assembly comprises a first magnet, the first magnetic gap is formed between the first magnet and a side wall of the magnetic bowl, the second magnet assembly comprises a second magnet and a third magnetic steel, the third magnetic steel is disposed on an outer side of the second magnet, and the third magnetic steel and the second magnet form the second magnetic gap.

6. The coaxial speaker box according to claim 5, wherein the first vibrating diaphragm comprises a first dome and a first folding ring, the first folding ring surrounds the first dome, the first folding ring comprises a first fixing portion, and the first fixing portion is attached and fixed to the inner wall surface of the frame.

7. The coaxial speaker box according to claim 6, wherein a second sound transmission hole is defined in a position, with respect to the first sound transmission hole of the frame, of the first fixing portion.

8. The coaxial speaker box according to claim 7, wherein parts of the first fixing portion forming the second sound transmission hole are attached and fixed to a bottom wall of the frame, where the first sound transmission hole is defined on the bottom wall of the frame.

9. The coaxial speaker box according to claim 6, wherein the first magnet assembly comprises a pole core, the pole core is disposed on the first magnet, the first fixing portion is partially clamped between an inner wall surface of the frame and the pole core.

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